

PROFILE

With a **passion for exploration** and a **strong ability to learn**, I hope **to discover my professional interests and acquire new skills** through the internship, better preparing myself for my future career.

EDUCATION

National Chung Cheng University

master of Communications Engineering - [NEAT Lab](#)

Sep 2024 - Present

National Chung Cheng University

bachelor of Communications Engineering

Sep 2020 - Jun 2024

WORK EXPERIENCE

National Chung Cheng University

Sep 2024 - Present

Teaching Assistant: **Embedded Operating Systems**

- Developed instructional materials for [FreeRTOS](#), provided technical support during lab sessions, and assisted students in resolving issues encountered in their experiments.

National Chung Cheng University

Feb 2024 - Jun 2024

Teaching Assistant: **Networked Embedded Systems**

- Reading academic papers and organizing the paper archive to support course materials.
- Organized lab meetings and managed daily tasks, demonstrating strong leadership.

AWARDS

Honorable Mention in National Chung Cheng University **Programming Competition** ([PDF](#))

2020

PROJECTS

[SAE J1772-Based EV and EVSE Simulator \(In Progress\)](#) ([GitHub](#)) | Skills: Embedded Systems, EV Technology, FreeRTOS, MQTT

- Developing an EV (Electric Vehicle) and EVSE (Electric Vehicle Supply Equipment) simulator based on the SAE J1772 standard using the ESP32 board and peripheral hardware to simulate the communication process & protocol between electric vehicles and charging stations.

Poker Card Image Classification (Kaggle Competition) ([GitHub](#)) | Skills: Machine Learning, CNN

- Participated in a Kaggle competition, using Convolutional Neural Networks (CNN) to classify card images.
- Ranked 5th out of 21 teams in class, achieving 94% accuracy.**

Bluetooth BLE-Based Smart Keyless Lock with Symmetric Encryption ([Video](#), [GitHub](#))

- This demo uses Bluetooth BLE and AES symmetric encryption to automatically unlock when within a specified range.

Innovative Communication Method Using Colors ([Video](#), [GitHub](#)) | Skills: Embedded Communication Interfaces, PWM

- Developed an **asynchronous communication** system that uses different colors to transmit information.
- The system is deployed on two ARM-based NUC140 development boards.
- Achieved 100% accuracy for transmitting 50+ characters.**

TECHNICAL SKILLS

Language: C/C++, Python, MATLAB

Frameworks/tools: PyTorch, TensorFlow, Git

Operation system: FreeRTOS, Linux

Embedded Systems: MCU, Eagle PCB, MQTT, Communication Interfaces (SPI, I²C, UART etc.)

COURSEWORK

- Microprocessor A
- Networked embedded system A
- Electric Circuits A+
- Electronics A
- Computer Networks A
- Python language & deep learning A+
- Machine learning A
- Probability A